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INDIAN INSTITUTE OF TECHNOLOGY GOA

भारतीय प्रौद्योगिकी संस्थान गोवा



CHENNAI
MATHEMATICAL
INSTITUTE

Deciding Conjugacy of a Rational Relation

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Rational Relations

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- A binary relation over monoids A^* and B^* is a subset of $A^* \times B^*$.
- A relation is *rational* if it can be built out of finite subsets of $A^* \times B^*$ using the operations union, product and Kleene star.

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- Example:

$\{(u, v) \mid u \in \{a, b, c\}^*, v \in \{b, c\}^* \text{ and } v \text{ is obtained from } u \text{ by replacing } a \text{ by } b\}$

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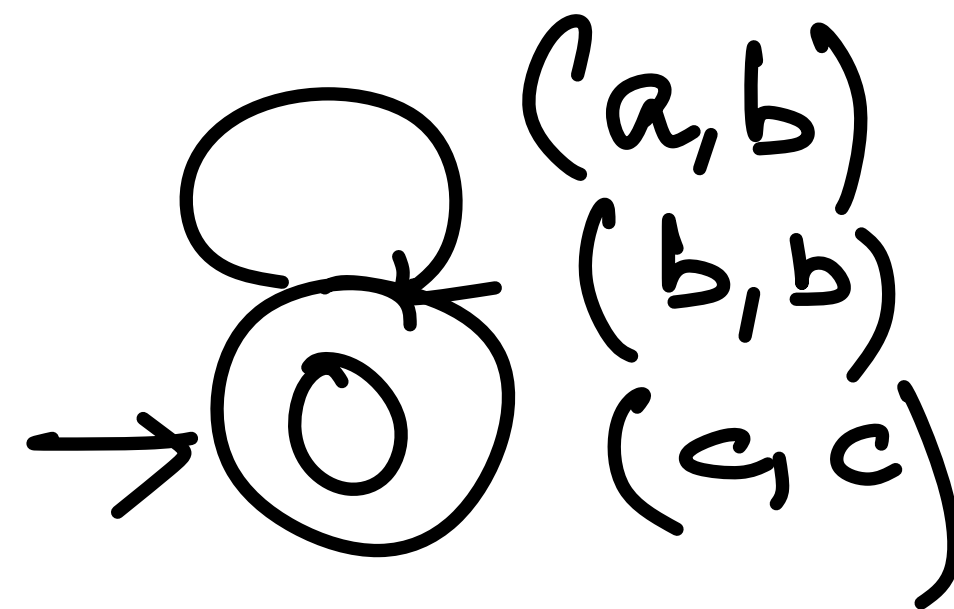
$\equiv$

$((a, b) + (b, b) + (c, c))^*$

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- Equivalently, rational relations are those accepted by finite state transducers (automata over pairs of words).
- Rational relations can model successive configurations of a Turing machine, instance of PCP ...

Instance of Post Correspondence Problem

- Given a collection of dominos tiles,

$$\left\{ \begin{bmatrix} t_1 \\ b_1 \end{bmatrix}, \begin{bmatrix} t_2 \\ b_2 \end{bmatrix}, \dots, \begin{bmatrix} t_k \\ b_k \end{bmatrix} \right\}$$

- PCP asks does there a sequence of dominos $i_1, i_2, \dots, i_l$ such that

$$t_{i_1} t_{i_2} \cdots t_{i_l} = b_{i_1} b_{i_2} \cdots b_{i_l}$$

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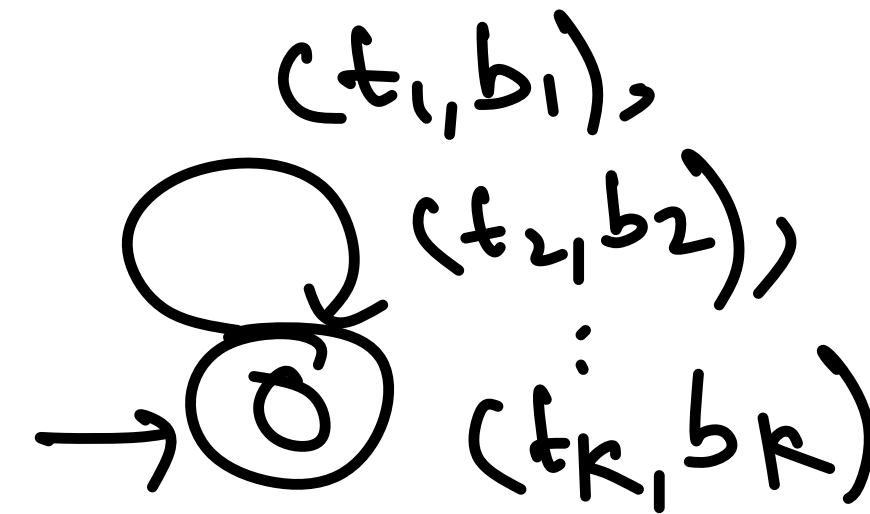
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- Modelled using a rational relation

$$R = ((t_1, b_1) + (t_2, b_2) + \dots + (t_k, b_k))^*$$



- Whether R contains an identical pair?

Instance of Post Correspondence Problem (Undecidable [Post'46])

- Given a collection of dominos tiles,

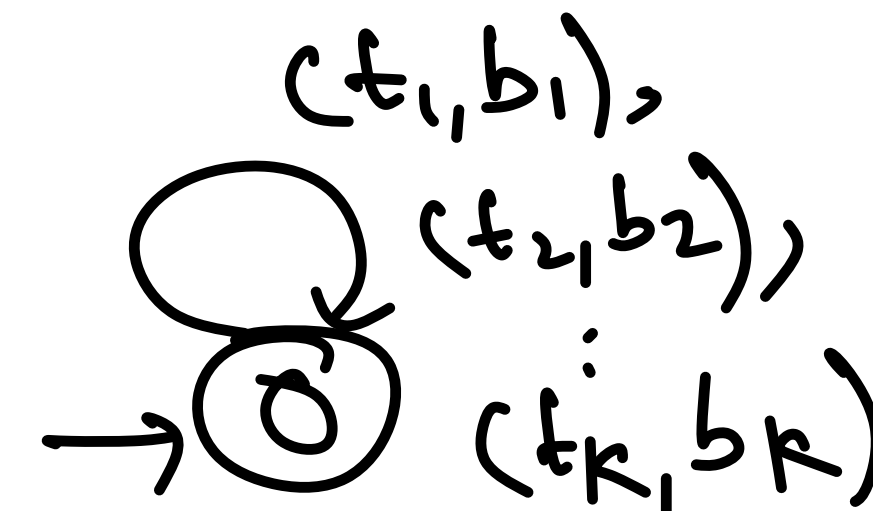
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- Whether R contains an identical pair?

Undecidable

Algorithmic Questions on Rational Relations

- Equivalence of two rational relation
- Disjointness of two rational relation
- Universality of a rational relation



Undecidable [Griffiths'68,
Fischer and Rosenberg'68,
Ibarra'78]

- Emptiness of a rational relation
- Finiteness of a rational relation



Decidable [Berstel'79]

- We study conjugacy of rational relations

Conjugate words

Two words u and v are conjugates

- if there exist words x and y such that

$$u = xy \text{ and } v = yx$$

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- if there exists a word z (witness) such that

$$uz = zv$$

[Lyndon-Schützenberger'62]

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listenlist

listenlist

- if there exists a word z (witness) such that

$$uz = zv \quad [\text{Lyndon-Schützenberger'62}]$$

Algorithmic Questions on Rational Relations

- Given a rational relation R
 - R contains an identical pair ? - Undecidable (Reduction from PCP)
 - R contains a conjugate pair ?

Instance of Conjugate PCP

- Given a collection of dominos tiles

$$\left\{ \begin{bmatrix} t_1 \\ b_1 \end{bmatrix}, \begin{bmatrix} t_2 \\ b_2 \end{bmatrix}, \dots, \begin{bmatrix} t_k \\ b_k \end{bmatrix} \right\}$$

- Conjugate PCP asks does there exist a sequence of dominos $i_1, i_2, \dots, i_l$ such that

$$t_{i_1} t_{i_2} \cdots t_{i_l} \curvearrowright b_{i_1} b_{i_2} \cdots b_{i_l}$$

- Modelled by a rational relation

$$R = ((t_1, b_1) + (t_2, b_2) + \cdots + (t_k, b_k))^*$$

- Whether R contains a conjugate pair?

Instance of Conjugate PCP (Undecidable [Halava-Harjit-Sahla' 21])

- Given a collection of dominos tiles

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- Modelled by a rational relation

$$R = ((t_1, b_1) + (t_2, b_2) + \dots + (t_k, b_k))^*$$

- Whether R contains a conjugate pair?

Undecidable

Algorithmic Questions on Rational Relations

- Given a rational relation R

Existence?

- R contains an identical pair ? - Undecidable (Reduction from PCP)
- R contains a conjugate pair ? - Undecidable (Reduction from conjugate PCP)

For all?

- R contains only identical pairs ?
- R contains only conjugate pairs ?

Instance of forall-variant of PCP

- Given a collection of dominos tiles,

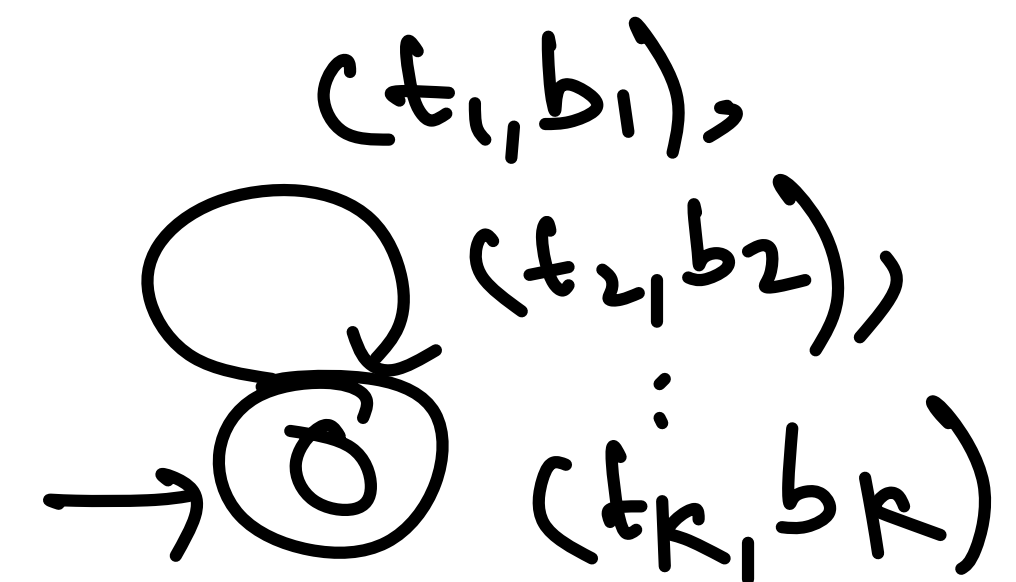
$$\left\{ \begin{bmatrix} t_1 \\ b_1 \end{bmatrix}, \begin{bmatrix} t_2 \\ b_2 \end{bmatrix}, \dots, \begin{bmatrix} t_k \\ b_k \end{bmatrix} \right\}$$

- Forall variant of PCP asks whether for every sequence of dominos $i_1, i_2, \dots, i_l$

$$t_{i_1} t_{i_2} \cdots t_{i_l} = b_{i_1} b_{i_2} \cdots b_{i_l}$$

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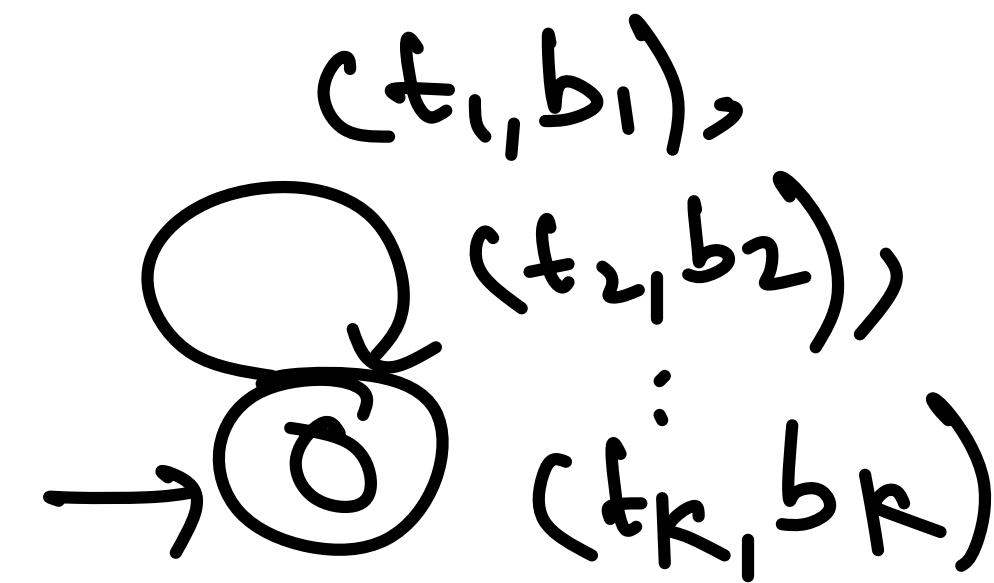
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- Forall variant of PCP asks whether for every sequence of dominos $i_1, i_2, \dots, i_l$

$$t_{i_1} t_{i_2} \dots t_{i_l} = b_{i_1} b_{i_2} \dots b_{i_l}$$

- Modelled using a rational relation

$$R = ((t_1, b_1) + (t_2, b_2) + \dots + (t_k, b_k))^*$$



- Whether R contains only identical pair?

Yes if $t_1 = b_1, t_2 = b_2, \dots, t_k = b_k$

No otherwise

Instance of forall-variant of Conjugate PCP

- Given a collection of dominos tiles,

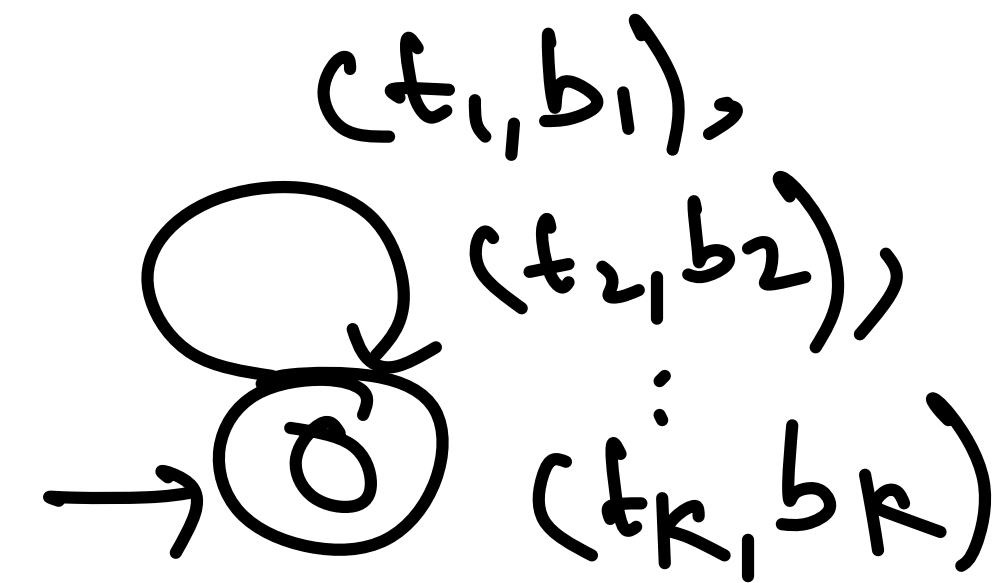
$$\left\{ \left[\frac{t_1}{b_1} \right], \left[\frac{t_2}{b_2} \right], \dots, \left[\frac{t_k}{b_k} \right] \right\}$$

- Forall variant of Conjugate PCP asks whether for every sequence of dominos $i_1, i_2, \dots, i_l$

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- Whether R contains only conjugate pair?

Instance of forall-variant of Conjugate PCP

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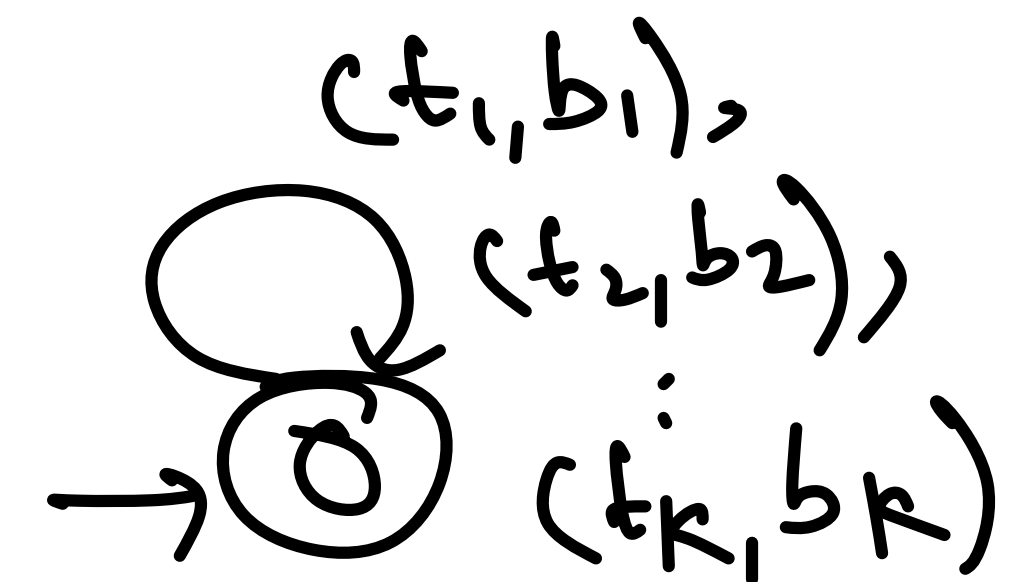
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- Forall variant of Conjugate PCP asks whether for every sequence of dominos $i_1, i_2, \dots, i_l$

$$t_{i_1} t_{i_2} \dots t_{i_l} \circ b_{i_1} b_{i_2} \dots b_{i_l}$$

- Modelled using a rational relation

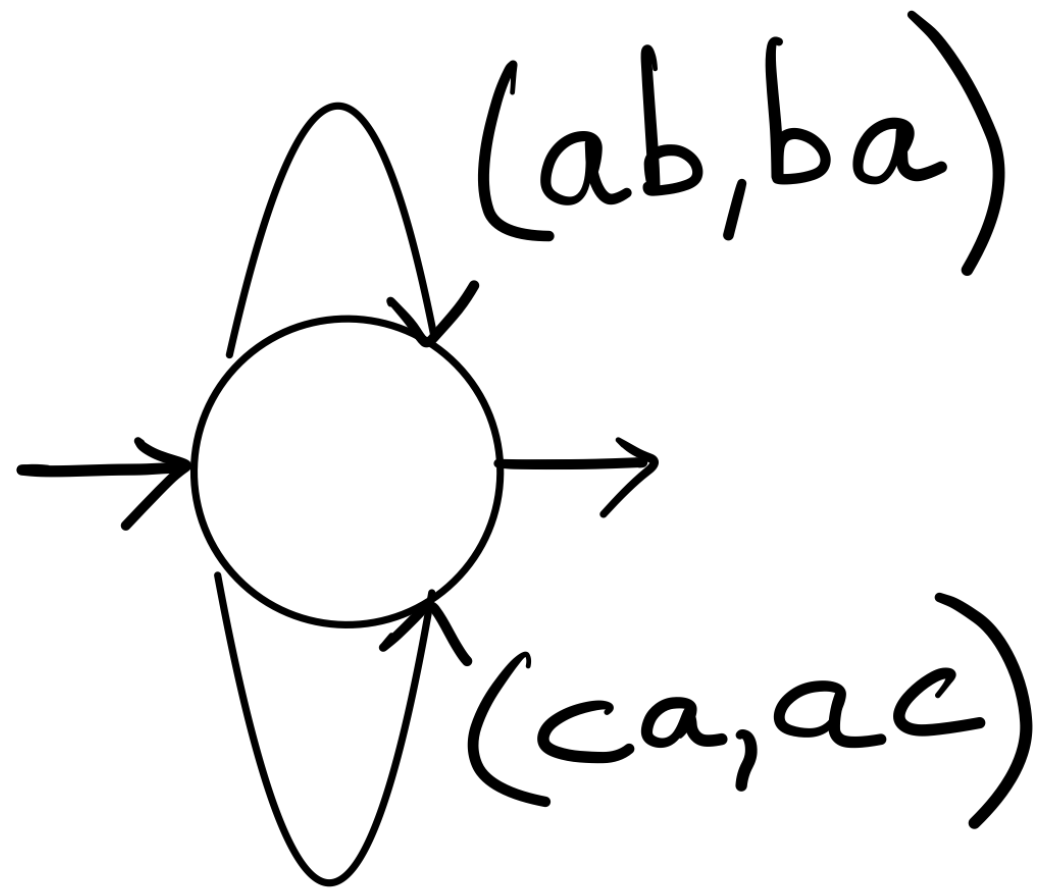
$$R = ((t_1, b_1) + (t_2, b_2) + \dots + (t_k, b_k))^*$$



- Whether R contains only conjugate pair?

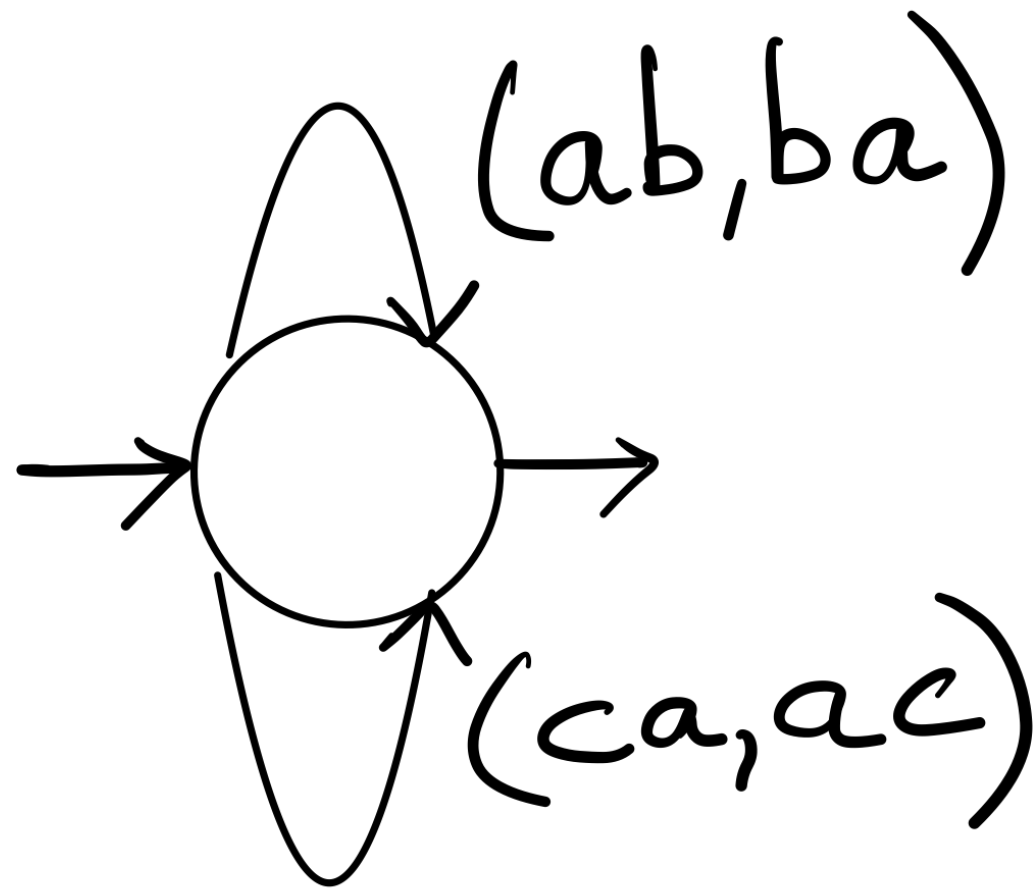
Is it enough if $t_1 \circ b_1, t_2 \circ b_2, \dots, t_k \circ b_k$?

More on conjugates



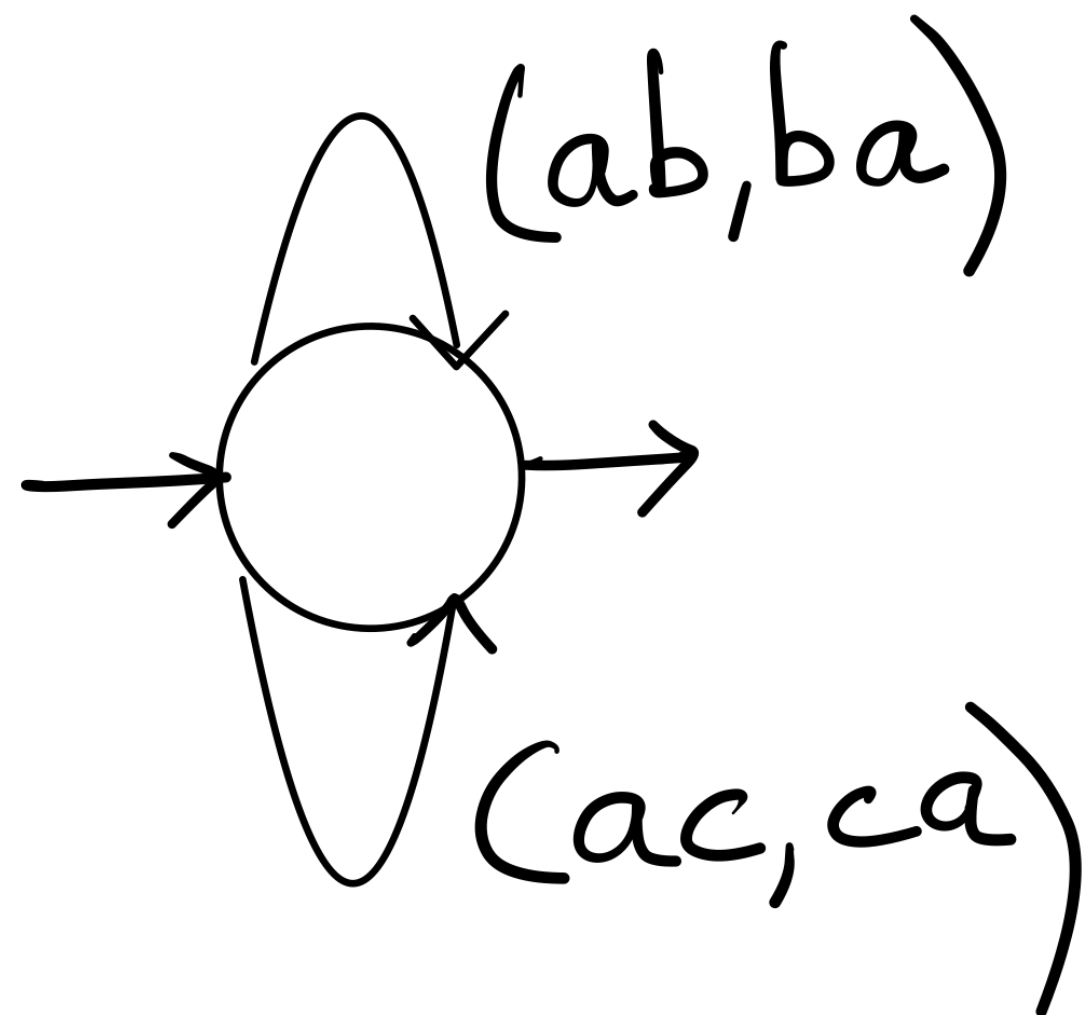
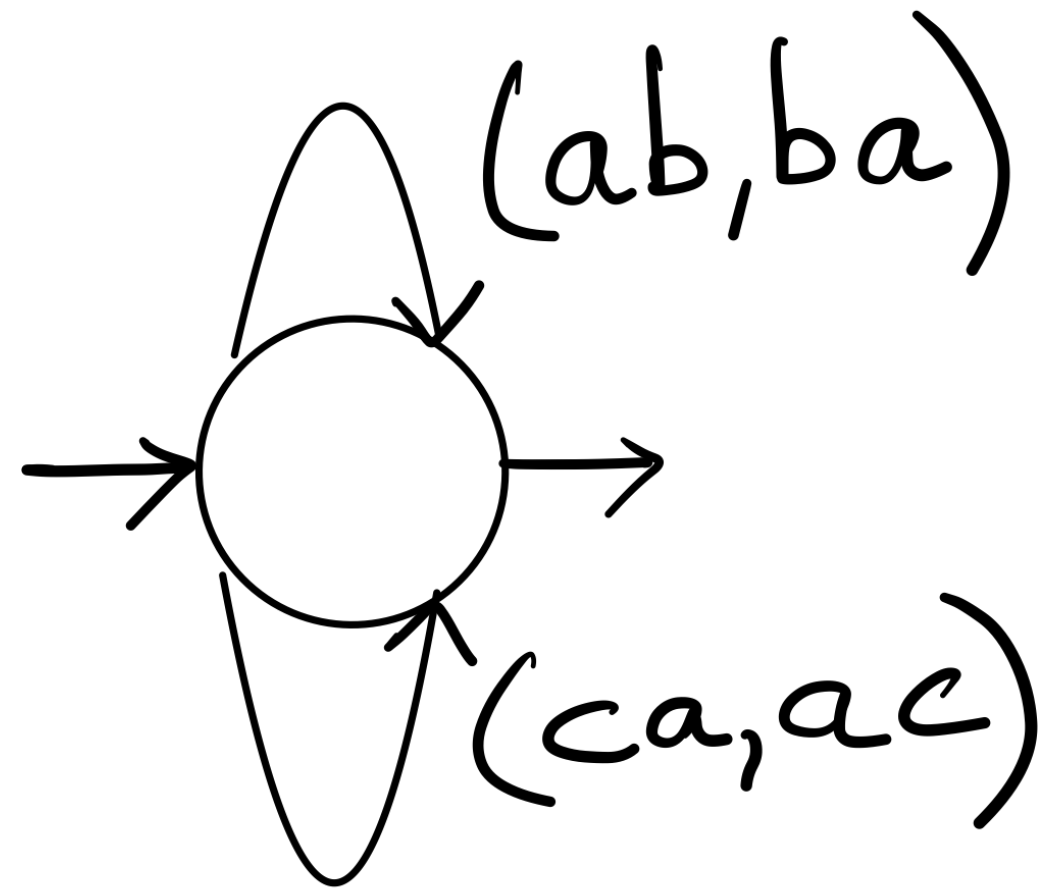
More on conjugates

$(abca, baac)$ is not conjugate



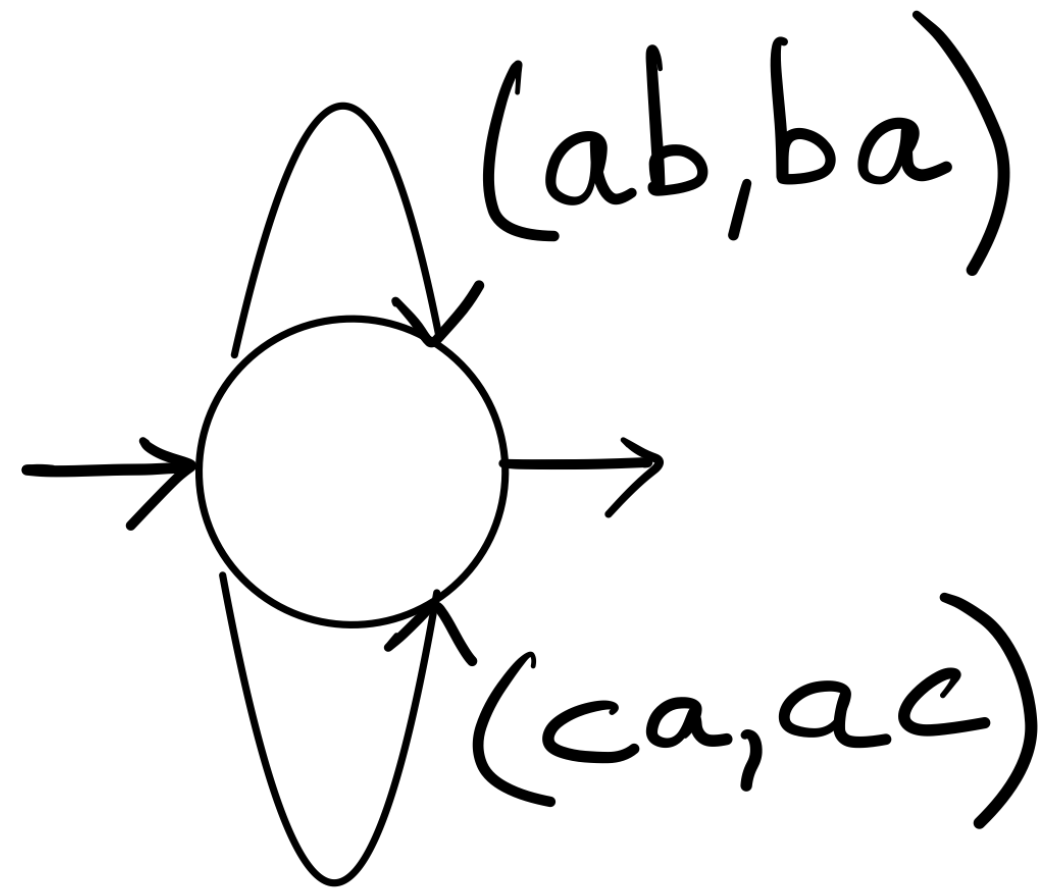
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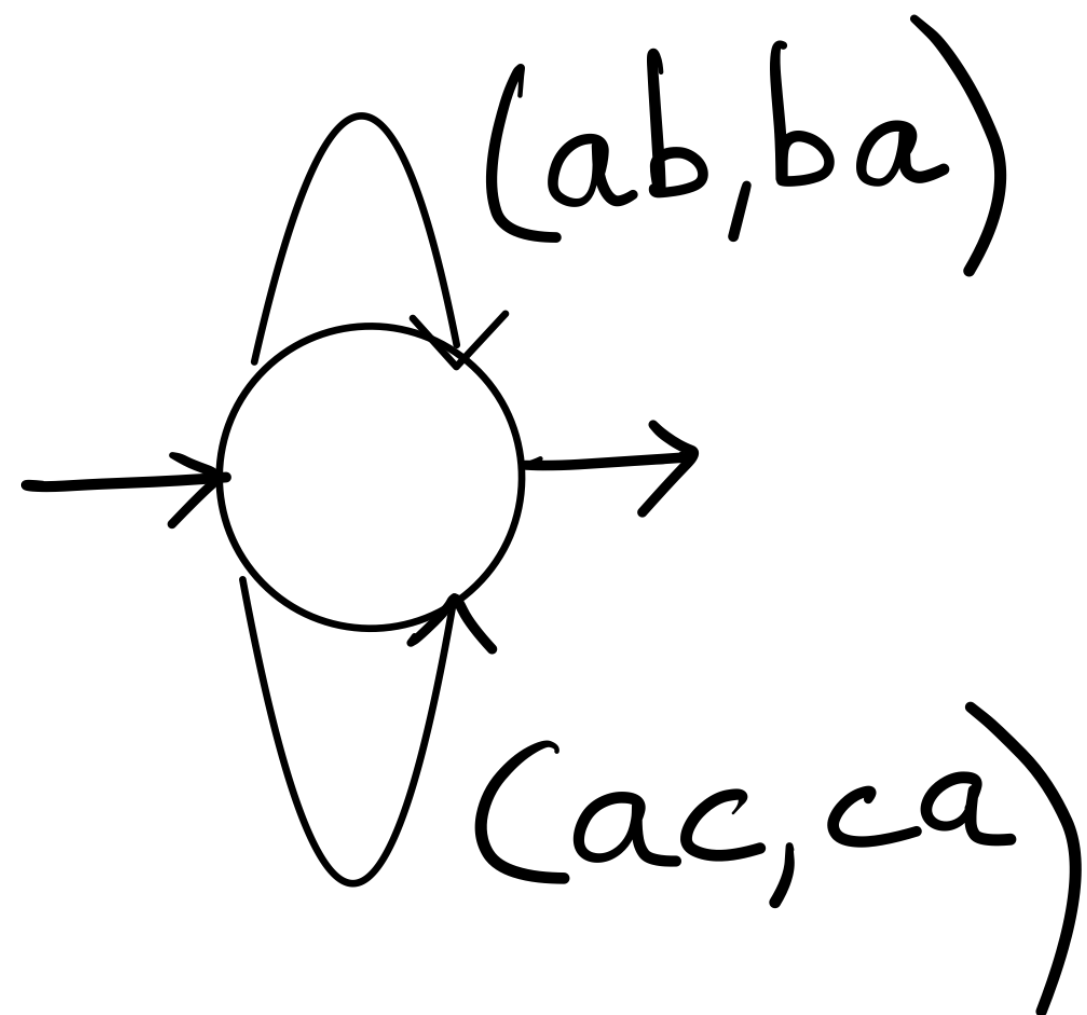


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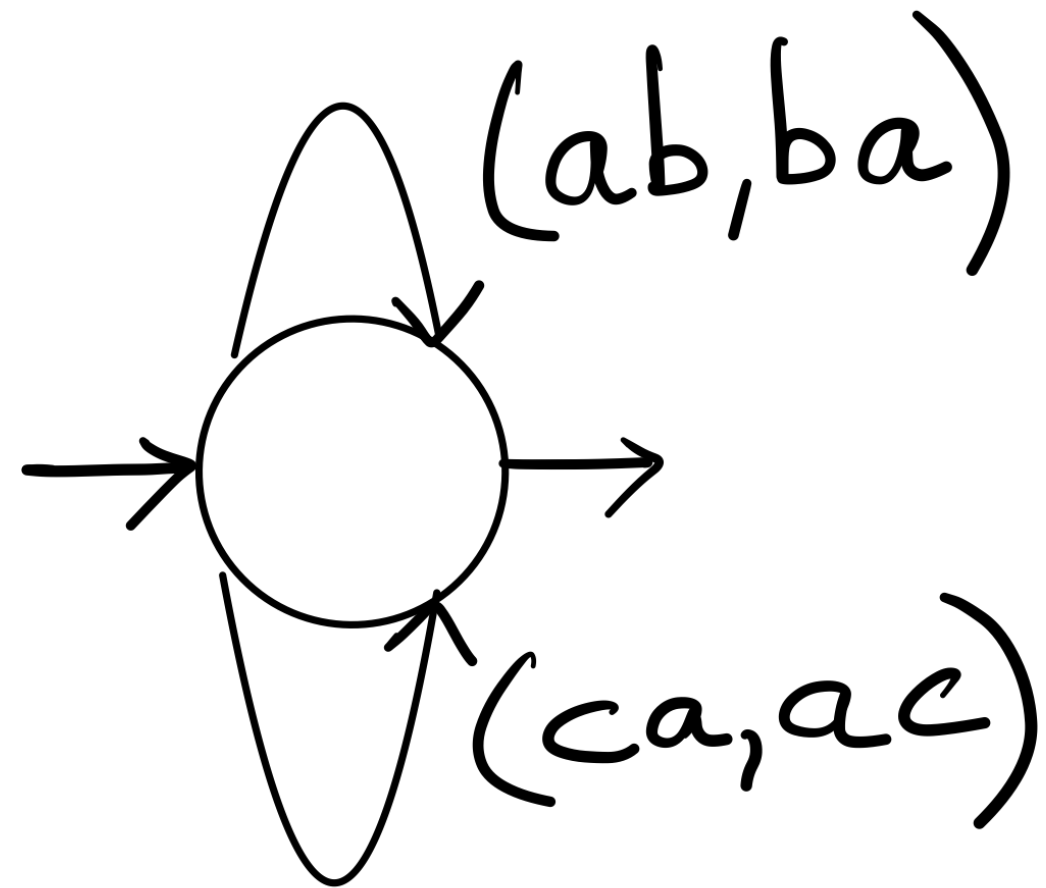


Any combination of pairs is conjugate

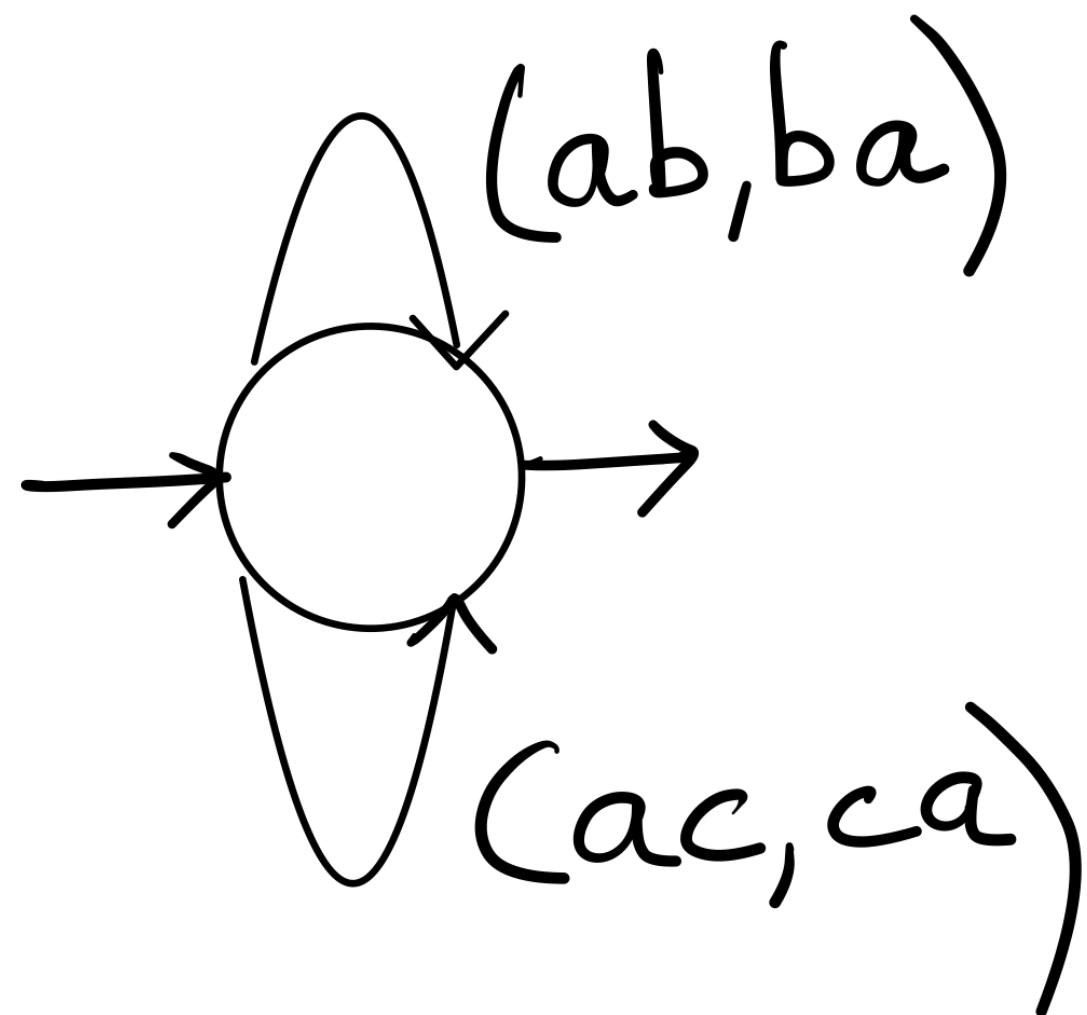


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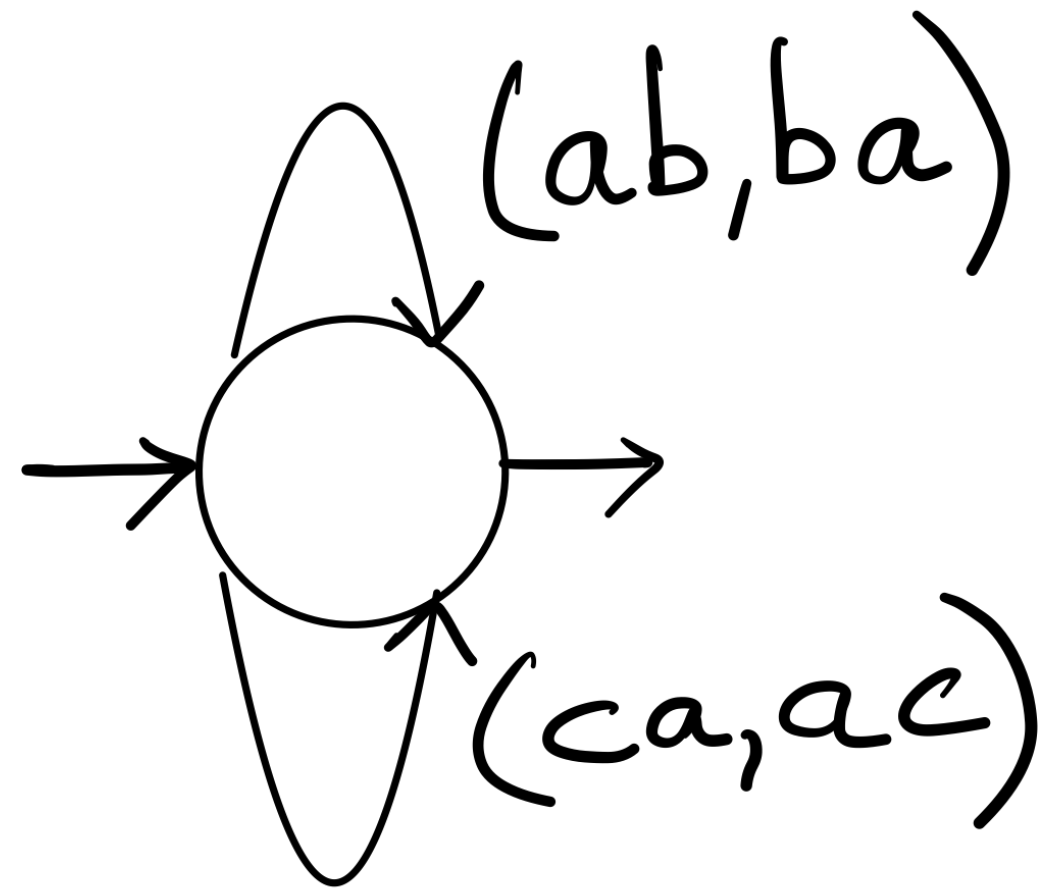
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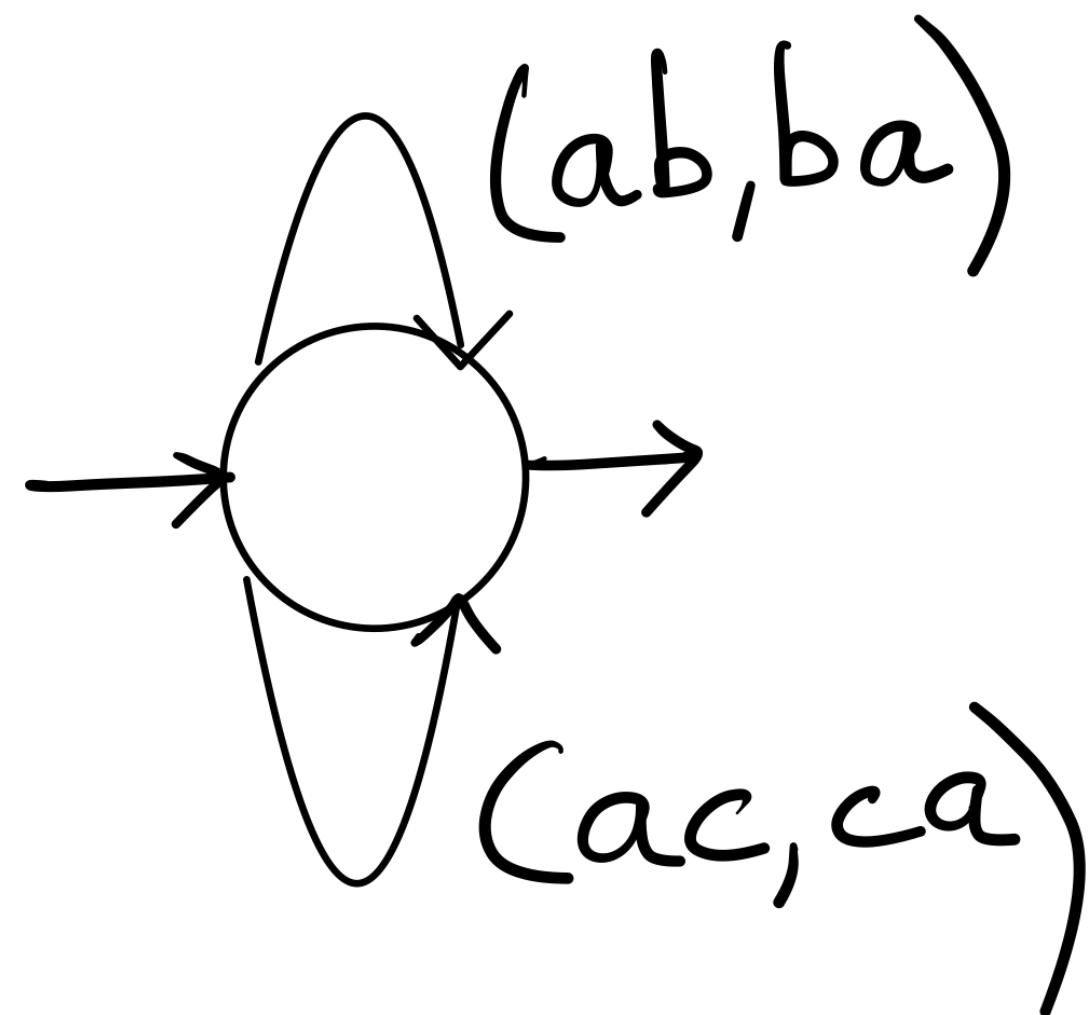
$$uz = zv$$

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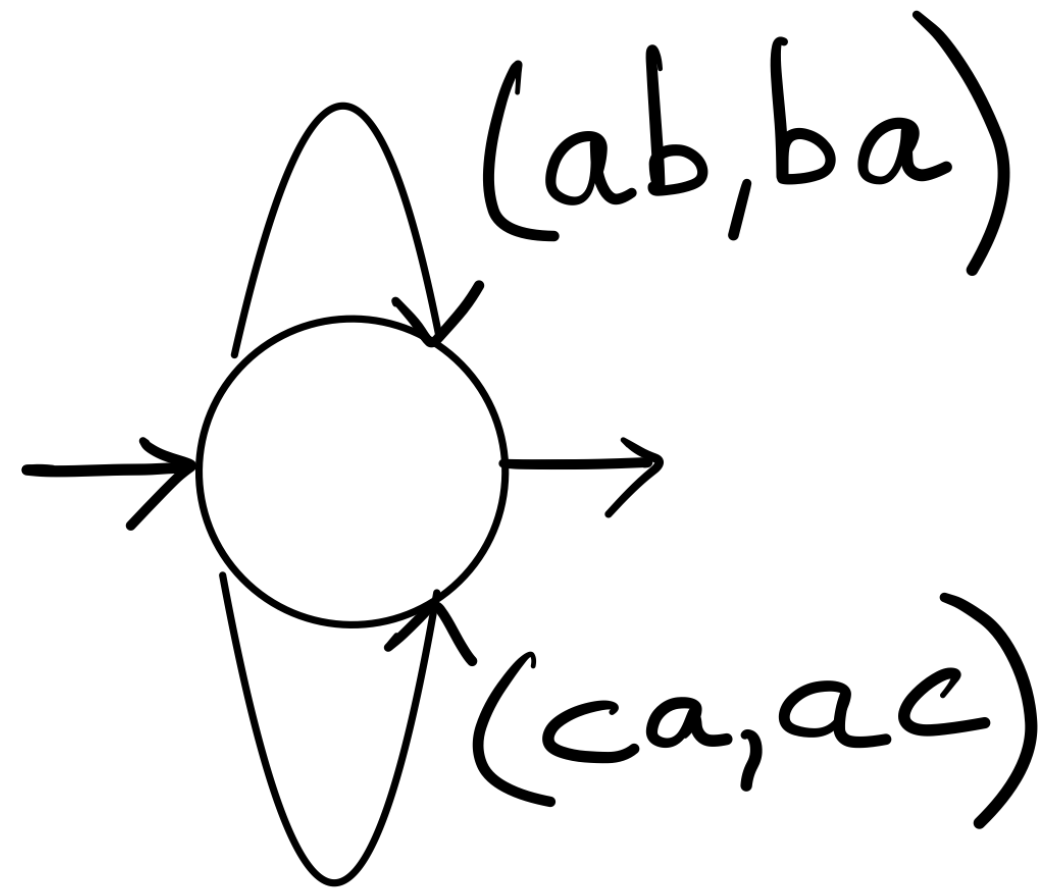
Any combination of pairs is conjugate


$$\begin{array}{l} a b \cdots a b \\ b a \cdots b a \end{array}$$

$$u z = z v$$

More on conjugates

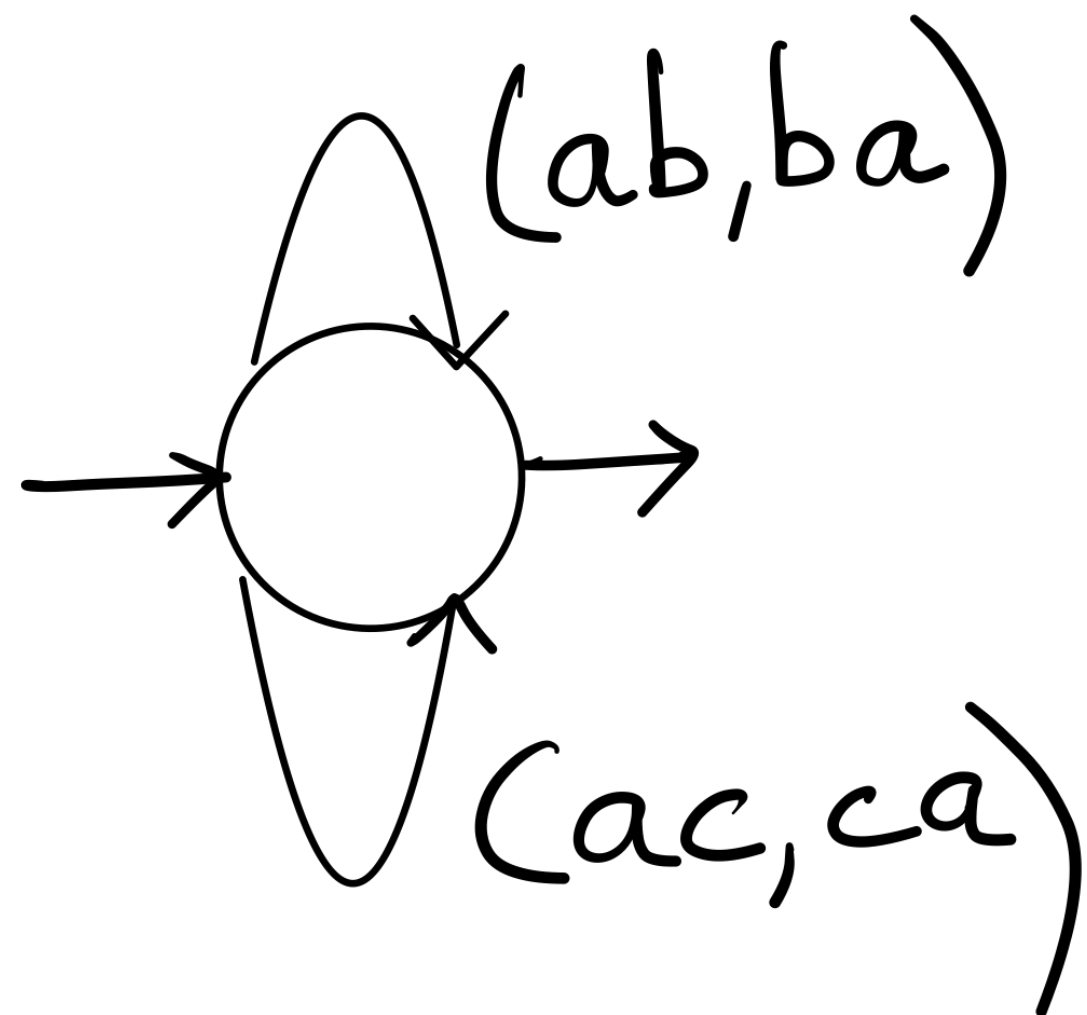
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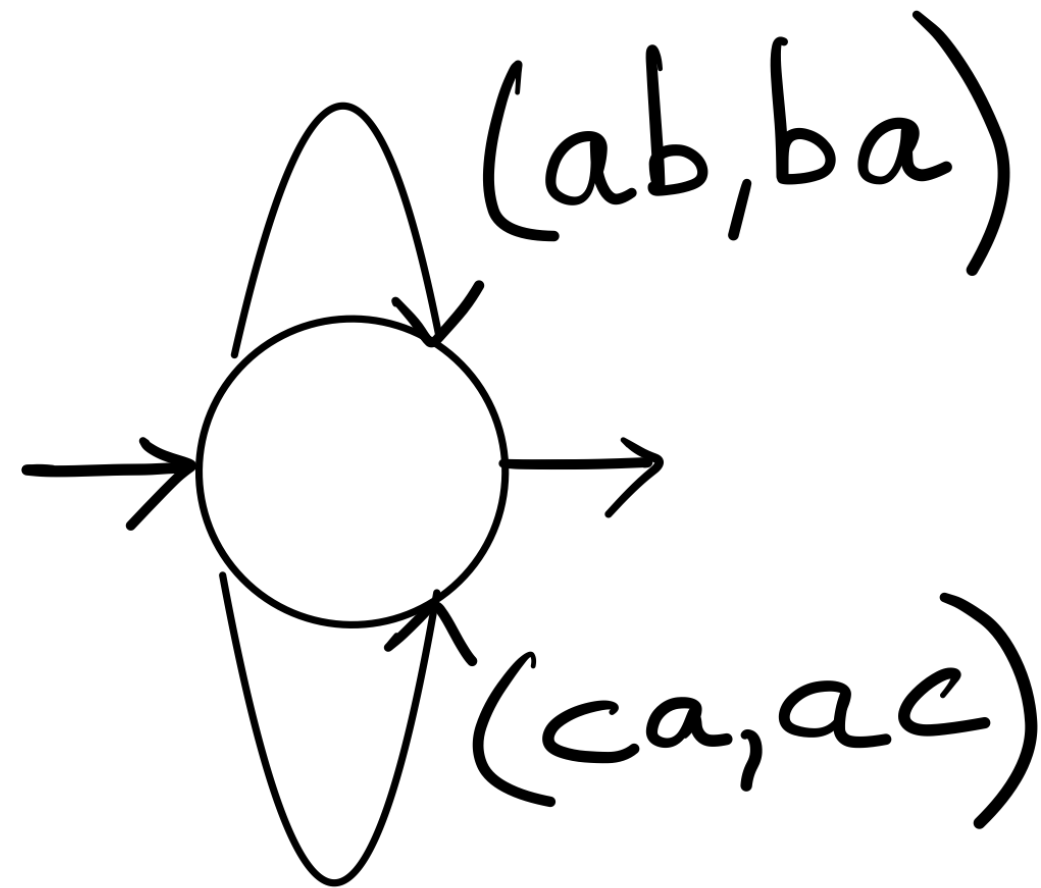
$$\begin{array}{c} a/b \cdots a/b \\ b/a \cdots b/a \end{array}$$

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More on conjugates

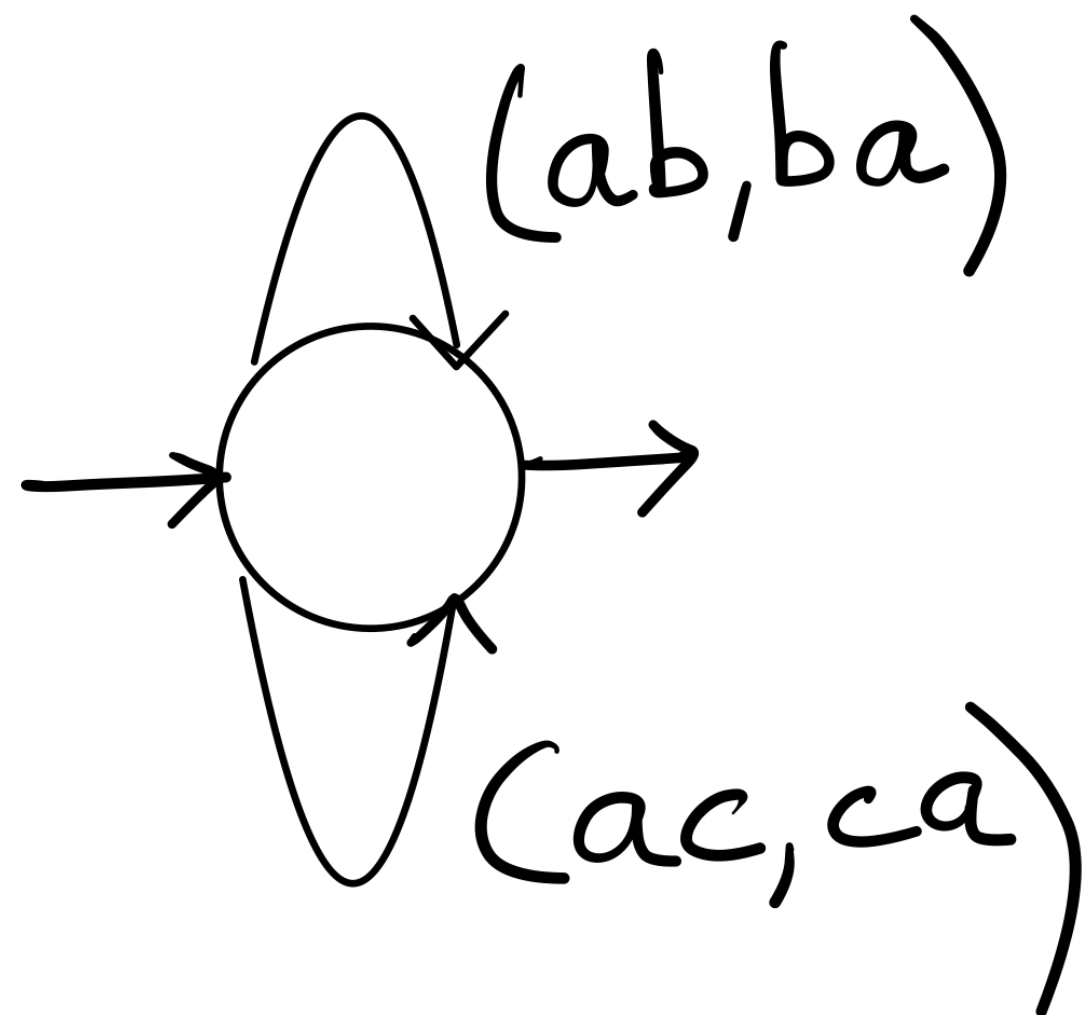
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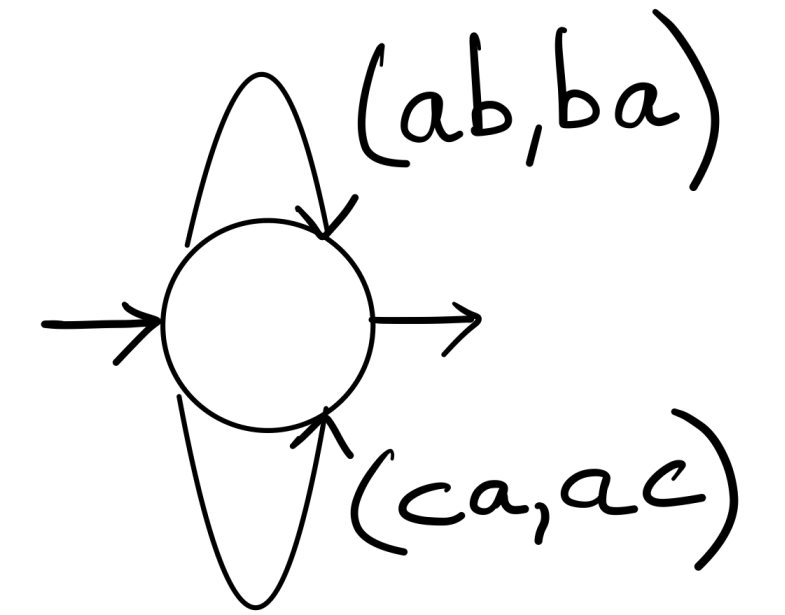
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$$\begin{array}{l} a/b \cdots a \ b/a/c \cdots a \ c/ \\ b \ a \cdots b/a/c \ a \cdots c/a \end{array}$$

$$uz = zv$$



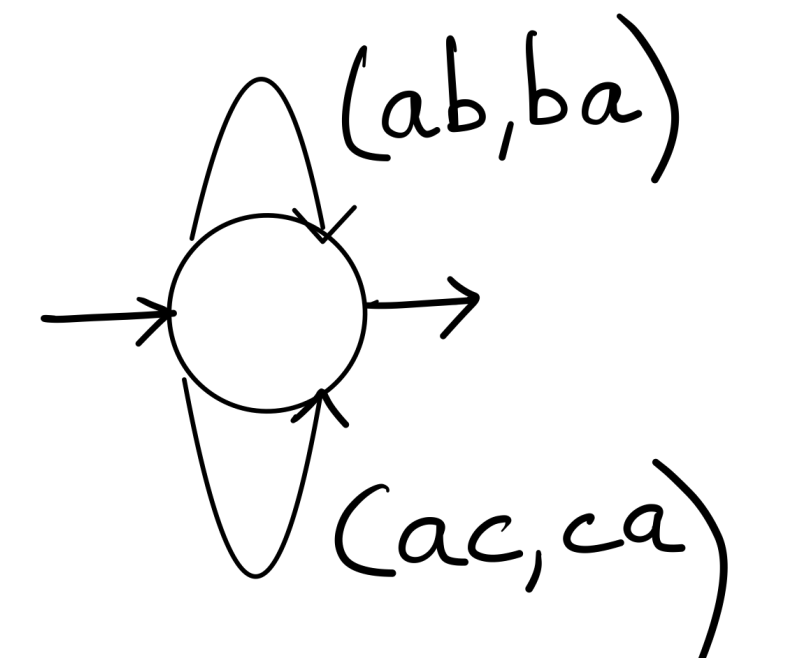
Conjugacy of Kleene star



G — set of pairs of words

$(abca, baac)$ is not conjugate

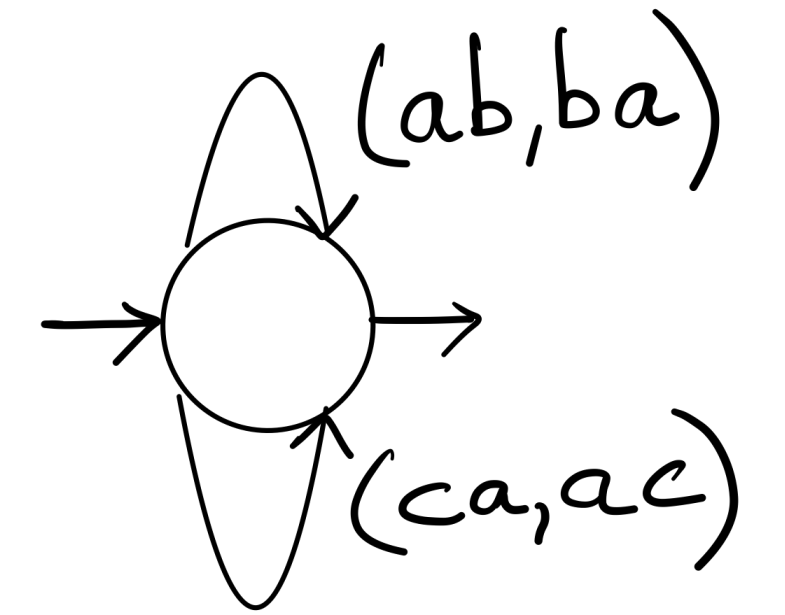
G^* — consist of pairs obtained by point wise concatenation of some pairs in G



Conjugate

$$\begin{array}{l} a/b \cdots a \quad b/a/c \cdots a \quad c/ \\ b \quad a \cdots b/a/c \quad a \cdots c/a \end{array}$$

Conjugacy of Kleene star

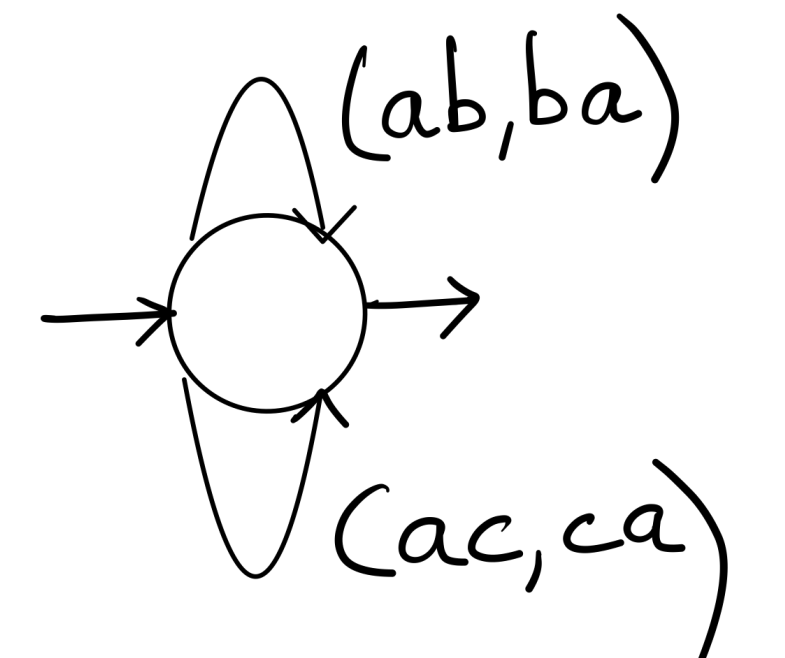


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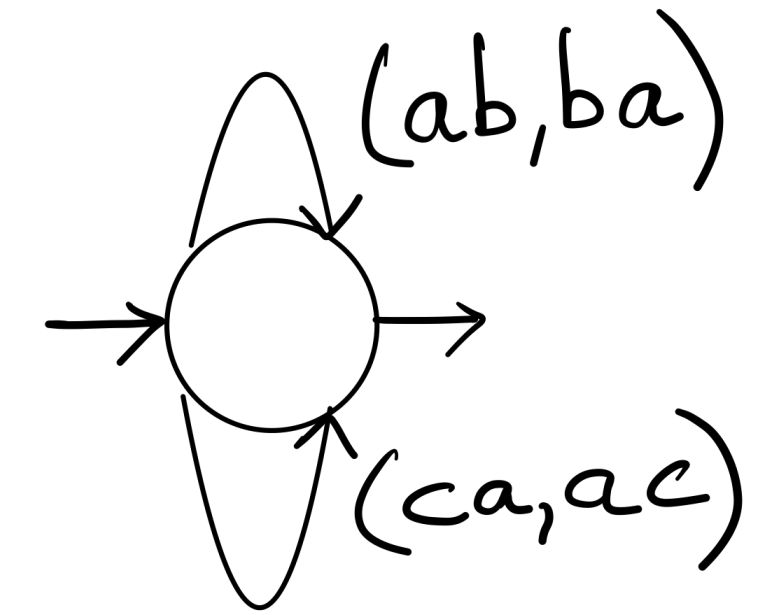
When is every pair in G^* conjugate?



Conjugate

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Conjugacy of Kleene star

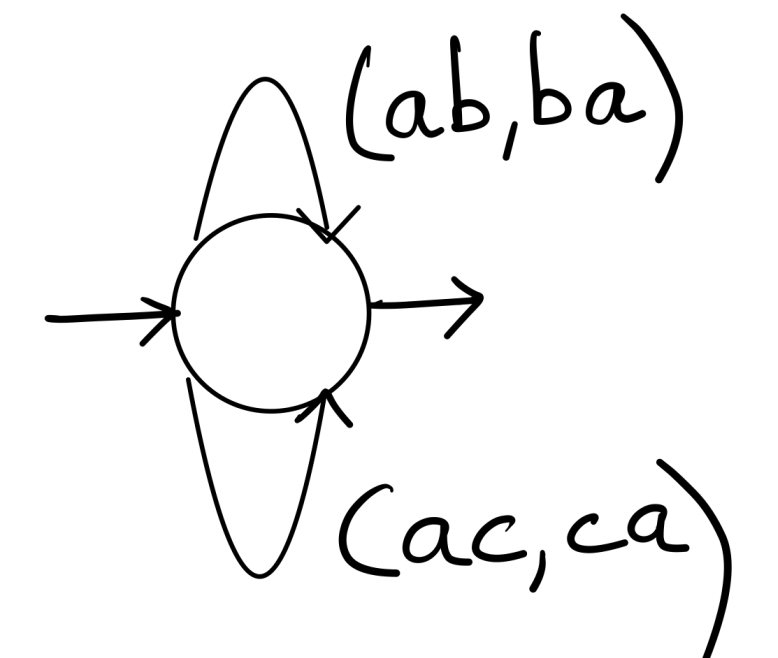


G — set of pairs of words

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When is every pair in G^* conjugate?



Conjugate

Theorem: The set G^* is conjugate iff G has a common witness

$a/b \cdots a \quad b/a/c \cdots a \quad c/$
 $b \quad a \cdots b/a/c \quad a \cdots c/a$

Common Witness Theorem of Kleene Closure

A word z is a common witness of set of pairs G if $\forall_{(u,v) \in G} \underbrace{uz = zv}_{\text{inner witness}}$ or $\forall_{(u,v) \in G} \underbrace{zu = vz}_{\text{outer witness}}$

Theorem:

Let G be a set of conjugate pairs. TFAE

1. G^* is conjugate.
2. G^* has a common witness z .
3. G has a common witness z .
4. Roots of G has a common witness z .

Example of Common witnesses

Consider set $P = \{(ab, ba), (ac, ca)\}$

(ab, ba) has inner witnesses $(ab)^*a$ and outer witnesses $(ba)^*b$.

(ac, ca) has inner witnesses $(ac)^*a$ and outer witnesses $(ca)^*c$.

P has one common inner witness $(ab)^*a \cap (ac)^*a = a$.

P does not have any common outer witness since $(ba)^*b \cap (ca)^*c = \emptyset$.

Example of Common witnesses

- The set $\{(ab, ba), (ac, ca)\}$ has a unique common witness a .
- The set $\{(ab, ba), (abab, baba)\}$ has infinitely many common inner witnesses $(ab)^*a$ and infinitely many common outer witnesses $(ba)^*b$.
- The set $\{(ab, ba), (ba, ab)\}$ has no common witness since $(ab)^*a \cap (ba)^*b = \emptyset$.

Number of Common witnesses

Proposition: Let G be a set of pairs. Then one of the following is true:

- G has no common witness.
- G has exactly one common witness.
- G have infinitely many common witnesses.

Common Witness Theorem of Kleene Closure

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Roots of a Set

- A word is *primitive* if it cannot be expressed as a power of a smaller word.

Example: *aba* is a primitive word but *abab* is not.

- *Primitive root* of a word u is the primitive word ρ_u such that $u = \rho_u^n$ for some $n \geq 1$.

Example: *ab* is the primitive root of *abab*.

Roots of a Set

- A word is *primitive* if it cannot be expressed as a power of a smaller word.
- *Primitive root* of a word u is the primitive word ρ_u such that $u = \rho_u^n$ for some $n \geq 1$.
- *Primitive root of a pair* (u, v) is (ρ_u, ρ_v) .
- *Roots of a set* P is the set of all primitive root of each pair in P .

Example: roots of the set $\{(abab, baba), (bab, abb)\}$ is $\{(ab, ba), (bab, abb)\}$.

Why roots are important?

- A primitive word cannot be equal to any of its nontrivial cyclic shifts.
- Let (u, v) be a pair of distinct conjugate primitive words

1. (u, v) has a unique cut.

(x, y) is a *cut* of (u, v) if $u = xy$ and $v = yx$

Why roots are important?

- A primitive word cannot be equal to any of its nontrivial cyclic shifts.
- Let (u, v) be a pair of distinct conjugate primitive words
 1. (u, v) has a unique cut (x, y) .
 2. Any cut of the pair (u^n, v^n) for $n \geq 1$ is of the form $((xy)^*x, (yx)^*y)$

Why roots are important?

- A primitive word cannot be equal to any of its nontrivial cyclic shifts.
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 1. (u, v) has a unique cut (x, y) .
 2. Any cut of the pair (u^n, v^n) for $n \geq 1$ is of the form $((xy)^*x, (yx)^*y)$

Consequence:

- z is a witness of (u, v) iff z is a witness of (ρ_u, ρ_v) .
- z is a witness of a set G iff z is a witness of roots of G .

Common Witness Theorem of Kleene Closure

A word z is a common witness of set of pairs G if $\forall_{(u,v) \in G} uz = zv$ or $\forall_{(u,v) \in G} zu = vz$

Theorem:

Let G be a set of conjugate pairs. TFAE

1. G^* is conjugate.
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We prove

$$4 \implies 3 \implies 2 \implies 1 \implies 4$$

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We prove

$$4 \implies 3 \implies 2 \implies (1 \implies 4)$$

G^* is conjugate $\implies$ Roots of G has a common witness

G^* is conjugate $\implies$ Roots of G has a common witness

- When G is a finite set of pairs - using case analysis

G^* is conjugate $\implies$ Roots of G has a common witness

- When G is a finite set of pairs - using case analysis
- When G is an infinite set of pairs -

Compactness Theorem: Let G be an infinite set of pairs. If every finite subset of G has a common witness, then G has a common witness.

G^* is conjugate $\rightarrow$ Closure of every finite subset G_f is conjugate $\rightarrow$ Roots of G_f has a common witness



By Compactness Theorem, Roots of G has a common witness

Conjugacy of Sumfree Set

Theorem: The set $(u_0, v_0)G_1^*(u_2, v_2)G_2^* \cdots G_k^*(u_k, v_k)$ is conjugate iff it has a common witness

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- This does not generalise to arbitrary sets.
- Example : $(ab, ba)^* + (ba, ab)^*$ is an infinite conjugate set but does not have a common witness.

Common Witness Theorem of Sumfree Set

A word z is a common witness of G if $\forall_{(u,v) \in G} uz = zv$ or $\forall_{(u,v) \in G} zu = vz$

Theorem: Let $M = (u_0, v_0)G_1^*(u_2, v_2)G_2^* \cdots G_k^*(u_k, v_k)$ be a sumfree set. TFAE

1. M is conjugate.
2. There exist a common witness z for each of the singleton reduxes (where all but one Kleene star is assumed to be ϵ)
3. M has a common witness z .

Conjugacy of a Rational Relation

- A set of pairs (or a relation) is conjugate if each pair in the set is conjugate.

Theorem: Conjugacy of rational relations is decidable

Deciding Conjugacy of a Rational Relation

1. Assume that the rational relation is given as a rational expression (E) over pairs.
2. The union operation of rational relation preserves conjugacy, while product and Kleene star do not.
3. Every rational expression is equivalent to a sum of sumfree expressions ($E \equiv e_1 + e_2 + \dots + e_k, k \geq 1$) where each e_i is sumfree.

$$E_1 + E_2 \equiv (e + \dots + f) + (g + \dots + h)$$

$$E^* \equiv (e + \dots + f)^* \equiv (e^* \dots f^*)^*$$

$$E_1 \cdot E_2 \equiv (e + \dots + f) \cdot (g + \dots + h) \equiv e \cdot g + \dots + f \cdot h$$

(Exponential blow up both in number of summands and size of individual summands)

Deciding Conjugacy of a Rational Relation

1. Assume that the rational relation is given as a rational expression (E).
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4. E is conjugate if each sumfree expressions are conjugate.
5. Check the conjugacy for each sumfree expression.

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4. E is conjugate if each sumfree expressions are conjugate.
5. Check the conjugacy for each sumfree expression.
 - a. Each sumfree expression is of the form $(u_0, v_0)E_1^*(u_1, v_1)E_2^*\cdots E_k^*(u_k, v_k)$, $k \geq 0$
 - b. $(u_0, v_0)E_1^*(u_1, v_1)E_2^*\cdots E_k^*(u_k, v_k)$, $k \geq 0$ is conjugate iff it has a common witness.
 - c. The common witnesses of a sumfree expression can be computed in $\mathcal{O}(h \cdot m^3)$ where h is the star height of the expression and m is the length of the expression.

Conjugacy of a Rational Relation

- *A set of pairs (or a relation) is conjugate* if each pair in the set is conjugate.

Theorem: Conjugacy of rational relations is decidable

- Applications:
 - Characterising the sequentiality of rational function (Choffrut '1977)
 - Edit distance of finite state transducers (Aiswarya-Manuel-S. '2024)

Thank you.